

EPC meeting #2 Fall 2020

Wednesday, 11/18/2020

10:00 – 12:00

Items will be discussed in the order that they were submitted.

Agenda:

1. BS in Chemical Engineering Proposed Changes
2. CHE 150 – new course
3. CHE 321 -
4. CHE 396 -
5. CHE 397 -
6. CHE 514 -
7. ECE Technical Elective Changes
8. BIOE 538 –
9. CS 532 –
10. CS 351 –
11. CS 378 –

Topics for discussion:

1. Start transferring control of EPC over to Miiri Kotche
2. Discuss and propose college-level criteria for instructional faculty qualifications in accordance with university policy
3. Discussion of potentially accepting credit for non-engineering 300/400 level courses:
 - a. MATH 310
 - b. STAT 361 / 362
 - c. FIN 300
 - d. MGMT 340, 350
 - e. MKTG 360
4. Credit hour overrides above 18 hours. Require DUGS permission. Close to 5% of College currently requests (~200)
5. Attempt to come up with a fixed schedule of EPC meetings for Spring 2021 and beyond

<i>Title:</i>	Revision of the Bachelor of Science in Chemical Engineering
<i>Sponsor:</i>	<i>Department of Chemical Engineering</i>
<i>Executive Summary:</i>	There are currently 18 required courses in the College of Engineering that comprise the BS in Chemical Engineering. One of these courses is CHE 396: Senior Design I. The department proposes to decrease the credit hours of this course from 4 to 3 hours to reflect the course load/level of effort required by students. In addition, changes to coursework in chemistry are reducing the hours of nonengineering coursework in the degree by two hours. Further, the department proposes to add a 3 credit hour course to the BS in Chemical Engineering degree requirement: CHE 230 Molecular Systems in Chemical Engineering. CHE 230 was approved as a new course in Spring 2020. As a result, the total hours required to earn the degree will remain at 128.
<i>Description:</i>	1. Decrease the credit hours for CHE 396: Senior Design I from 4 to 3 hours. 2. The credit hours of CHEM 232: Organic Chemistry I and CHEM 234 Organic Chemistry II are being reduced by one hour each (from 4 to 3 hours). As a result, the hours of nonengineering and General Education coursework in the degree will decrease by two hours (from 70 to 68 hours). 3. Add CHE 230 Molecular Systems in Chemical Engineering (3 credit hours) to the BS in Chemical Engineering degree requirement. These changes result in a change to the Total Hours Required in College of Engineering from 49 to 51. The total hours required to earn the degree will remain at 128.
<i>Justification:</i>	1. CHE 396 is the first course of the two semester capstone chemical engineering senior design course sequence (CHE 396/397); recent improvements in the course have decreased the course load and justifies a 3-credit hour course. 2. CHE 230 is designed to provide an introduction to molecular engineering and material science in relation to chemical engineering. It covers both materials chemistry and physics of bulk materials and nanomaterials for a broad range of applications and is an important addition in keeping the chemical engineering curriculum up-to-date. 3. Due to the change in CHE396 credit hours, addition of CHE 230, and change in CHEM 232 and CHEM 234 credit hours, the Total Hours Required in COE needs to be increased and the Non-Eng/GenEd requirements need to be decreased to reflect this change.

	4. Due to approved changes in CHEM courses from LAS, the undergraduate catalog needs to be revised to reflect these changes.
<i>Catalog Statement:</i>	<i>See below</i>
<i>Minority Impact Statement:</i>	<i>The changes will clarify the degree requirements for all students and will not adversely impact minority students.</i>
<i>Budgetary and Staff Implications:</i>	N/A
<i>Library Resource Implications:</i>	N/A
<i>Space Implications:</i>	N/A
<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	Provide unit dates, including department, college/school committees, faculty Approvals, as applicable. Chemical engineering faculty approval: Chemical Engineering department approval: Education Policy Committee approval:
<i>Contact Person:</i>	Alan Zdunek zdunek@uic.edu
<i>Proposed Effective Date/Term:</i>	Fall 2021

Current	Revised
BS in Chemical Engineering <i>Degree Requirements</i> To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students	BS in Chemical Engineering SAME

should consult the *College of Engineering* section for additional degree requirements and college academic policies.

BS in Chemical Engineering Degree Requirements Hours	
Nonengineering and General Education Requirements	70
Required in the College of Engineering	49
Selective ^a	3
Technical Elective ^a	3
Electives outside the Major Rubric ^a	3
Total Hours—BS in Chemical Engineering	128

^a Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.

Nonengineering and General Education Requirements

ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3
Exploring World Cultures course ^a	3
Understanding the Creative Arts course ^a	3
Understanding the Past course ^a	3
Understanding the Individual and Society course ^a	3
Understanding U.S. Society course ^a	3
MATH 180—Calculus I ^b	4
MATH 181—Calculus II ^b	4
MATH 210—Calculus III ^b	3
MATH 220—Introduction to Differential Equations	3
PHYS 141—General Physics I (Mechanics) ^b	4
PHYS 142—General Physics II (Electricity and Magnetism) ^b	4
CHEM 122- General College Chemistry I ^b (Lecture)	4
CHEM 123- General College Chemistry I ^b (Laboratory)	1
<u>OR</u> CHEM 116 – Honors and Major General and Analytical Chemistry I	5
CHEM 124—General College Chemistry II ^b (Lecture)	4
CHEM 125—General College Chemistry II ^b (Lab)	1
<u>OR</u> CHEM 118 – Honors and Major General and Analytical Chemistry II	5
CHEM 222—Analytical Chemistry ^c	4
CHEM 232—Organic Chemistry I	4
CHEM 233- Organic Chemistry Laboratory	2
CHEM 234—Organic Chemistry II	4
CHEM 342—Physical Chemistry I	3
CHEM 346—Physical Chemistry II	3
Total Hours- Nonengineering and General Education Requirements	70

^a Students should consult the General Education section of the catalog for a list of approved courses in this category.

^b This course is approved for the Analyzing the Natural World General Education category.

^c Students who take CHEM 116 and CHEM 118 to fulfill the general chemistry requirement do not need to take CHEM 222. Instead they should enroll in one of the following: CHEM 314: Inorganic Chemistry; CHEM 452: Biochemistry; or CHEM 402:

BS in Chemical Engineering Degree Requirements Hours	
Nonengineering and General Education Requirements	68
Required in the College of Engineering	51
Selective	3
Technical Elective ^a	3
Electives outside the Major Rubric ^a	3
Total Hours—BS in Chemical Engineering	128

^a Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.

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CHEM 232—Organic Chemistry I	3
CHEM 233- Organic Chemistry Laboratory	2
CHEM 234—Organic Chemistry II	3
CHEM 342—Physical Chemistry I	3
CHEM 346—Physical Chemistry II	3
Total Hours- Nonengineering and General Education Requirements	68

^a Students should consult the General Education section of the catalog for a list of approved courses in this category.

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CHE 451 Renewable Energy Technologies (3) CHE 453 Fundamentals of Electrochemistry (3) CHE 454 Molecular and Macromolecular Engineering (3) CHE 455 Nanoscale Systems in ChE (3) CHE 456—Fundamentals and Design of Microelectronics Processes (3) CHE 457 Colloidal and Interfacial Phenomena (3) CHE 494—Selected Topics in Chemical Engineering (3) Total Hours—Technical Elective a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee. b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering.	3 SAME
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<i>Title:</i>	Addition of a New General Education Course titled "Climate Engineering for Global Warming"
<i>Sponsor:</i>	<i>Department of Chemical Engineering</i>
<i>Executive Summary:</i>	The department is proposing to add a new general education course, CHE 150, Titled "Climate Engineering for Global Warming". This course will be placed under the General-Education Core category of "Analyzing the Natural World" and "Understanding the Past". It will be offered to undergraduate students as a 3 credit hour course and will be open to students from all majors.
<i>Description:</i>	This addition of the course does not affect the total credit hours required for the program.
<i>Justification:</i>	Climate change is a topic of immense and imminent importance. In this regards, this course is designed to help students understand the historical and engineering perspective of climate change, strategies to engineer earth's climate, monitor and predict climate change, and policies to mitigate risks associated with climate change. Through this course, the students will develop an understanding of the various aspects of science, engineering, economics, and policies related to climate change. This course will answer the following questions: 1. How can climate engineering impact global warming and climate change? 2. What is the science behind carbon dioxide removal and albedo modification? 3. What (and how) renewable technologies contribute to decarbonization? 4. How much time will the climate engineering strategies take to impact climate change on the earth? 5. How can science, policies, and discussions inform choices to reduce the risks posed by climate change? Since this is an important topic that the ChE department is addressing, the department wishes to offer the course permanently under its own rubric, number and name. The ChE department has had several discussions and meetings with the Earth and Environmental Sciences (EAES) department regarding a similar course, EAES 101 Global Environmental Change. After several discussions, we have received approval to offer CHE 150 from the EAES department. Both departments feel it is a good addition to the General Education course list and will compliment EAES 101 course by focusing on the engineering perspective of climate change.
<i>Catalog Statement:</i>	CHE 150. Climate Engineering for Global Warming. 3 hours. Overview of the engineering perspective of climate change and strategies to engineer the earth's climate and mitigate risks associate with global warming. There is no change in the BS program in Chemical Engineering

<i>Minority Impact Statement:</i>	<i>The changes will clarify the degree requirements for all students and will not adversely impact minority students.</i>
<i>Budgetary and Staff Implications:</i>	<i>None expected</i>
<i>Library Resource Implications:</i>	<i>None expected</i>
<i>Space Implications:</i>	<i>None expected</i>
<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	Provide unit dates, including department, college/school committees, faculty Approvals, as applicable. Chemical engineering faculty approval: 01/24/20 Chemical Engineering department approval: 01/24/20 Education Policy Committee approval:
<i>Contact Person:</i>	Alan Zdunek zdunek@uic.edu
<i>Proposed Effective Date/Term:</i>	Fall 2021

There is no change in the BS program in Chemical Engineering

Attached: CHE 150 Syllabus

Syllabus

CHE 150: Climate Engineering for Global Warming

Semester: Fall or Spring

Credit Hours: 3.00

Instructor: Meenesh Singh

Purpose of the Course

This is a general education course under the General-Education Core category of “Analyzing the Natural World” and “Understanding the Past”.

Historically, the earth’s climate has been changing since the earliest of the five major ice ages (more than 2 billion years ago) to the current greenhouse age. The earth has gone from having a methane atmosphere (greenhouse gas) to having excessive oxygen in the atmosphere, presumably from bacterial respiration. The influence of the species and earth causes its climate to change. Just what is climate? Climate is commonly thought of as the expected weather conditions at a given location over time (from Ice-age). People know when they go to New York City in winter, they should take a coat. When they visit the Pacific Northwest, they take an umbrella. Climate can be measured at many geographic and temporal scales—for example, cities, countries, or the entire globe—by such statistics as average temperatures, the average number of rainy days, and the frequency of droughts. *Climate change refers to changes in these statistics over the years, decades, and centuries.*

Climate engineering can be defined as the deliberate large-scale modification of the earth’s climate systems to counteract and mitigate anthropogenic climate change. The strategies which fall under this definition are loosely organized into four types: Decarbonizing the Energy Systems, Carbon Dioxide and other Green House Gas (GHG) Removal, Solar Radiation Management (Albedo Modification), and Adaptation to Climate Change. This course is designed to help students understand the historical and engineering perspective of climate change, strategies to engineer earth’s climate, monitor and predict climate change, and policies to mitigate risks associated with climate change. Through this course, the students will develop an understanding of the various aspects of science, engineering, economics, and policies of climate change. This course will answer the following questions:

6. How can climate engineering impact global warming and climate change?
7. What is the science behind carbon dioxide removal and albedo modification?
8. What (and how) renewable technologies contribute to decarbonization?
9. How much time will the climate engineering strategies take to impact climate change on the earth?
10. How science, policies, and discussions can inform choices to reduce the risks posed by climate change?

Recommended Texts

1) John Houghton, Global Warming: The Complete Briefing, 5th Edition, 2015, Cambridge Univ. Press.

2) D. Jacob, Introduction to Atmospheric Chemistry

<http://acmg.seas.harvard.edu/people/faculty/djj/book/index.html>

3) Edmond A. Mathez, and Jason E. Smerdon, Climate Change: The Science of Global Warming and our Energy Future, 2nd Edition, 2018, Columbia University Press, NY

Course Outline

The course will have the following five units and associated lectures:

Unit 1: The History of Climate Change and Introduction to Climate Engineering

1. History of anthropogenic climate change
2. Atmosphere chemistry and greenhouse gas (GHG) emissions
3. Carbon cycle and acidification of oceans
4. Introduction to climate engineering and strategies

Unit 2: Climate Monitoring and Prediction

5. Monitoring climate change using satellites
6. Monitoring of CO₂ and GHG levels in the atmosphere
7. Prediction of climate change – types of climate models
8. Prediction of global warming – Sawyer model, Broecker model, Hansen Model
9. Prediction of global warming – IPCC's models.

Unit 3: Solar Radiation Management (SRM)

10. Earth's thermostat and radiation balance
11. Greenhouse chemistry and the greenhouse effect
12. Radiative forcing and feedback of the climate system
13. SRM strategies – space reflectors and stratospheric aerosols
14. SRM strategies – cloud whitening, albedo modification
15. SRM strategies - other geoengineering techniques

Unit 4: Decarbonization of Energy Systems and GHG Removal

16. Decarbonization - Effective utilization of coal and natural gas
17. Decarbonization – Implementation of nuclear power
18. Decarbonization – Utilizing wind and solar power
19. Decarbonization – Harnessing hydropower
20. Decarbonization – Electrifying transportation with batteries and fuel cells
21. Introduction to carbon dioxide and other GHG removal
22. Carbon capture techniques – Post-combustion, oxyfuel, and pre-combustion
23. Carbon capture from air and seawater
24. Carbon sequestration – mineralization and deep sea injection
25. Carbon recycling – Renewable and green chemicals

Unit 5: Adapting to Climate Change, its Economics, and Policies

26. Adapting to changing ecosystem – deforestation, agriculture and food
27. International climate policy and emissions trading
28. Implementation of climate policy – Kyoto Protocol, international carbon action partnership
29. Economics of climate change – carbon tax, regulating emissions
30. Preparing for the future

Teaching and Learning Methods

A variety of teaching and learning approaches will be used in this course. Each unit will be introduced with a case study. Classes will vary to suit the topic under study and will include a mix of lectures, demonstrations and group based workshops. All classes will be organized to maximize interaction between the instructor and the students. The emphasis will be on developing a strong understanding of scientific basics and tools for analysis, rather than a descriptive presentation of processes.

Assessment

Assessment Item	Date	Weighting
Unit Evaluation	Continuous	15%
Exam on Unit 1-2	September	20%
Exam on Unit 3-4	October	20%
Team project	Start: October Submit: December	30%
Final Exam	Exam week	20%

Unit Evaluation: A set of tutorial problems will accompany each unit. These are to be done as homework. Class problem solving sessions are timetabled to help students with these tasks. Students may hand in problem solutions for feedback. Worked solutions will be provided.

Exams: The exams will be open book. Questions will predominantly test students' understanding of the topics covered in classroom. The first two exams will be on Unit 1-2 and Unit 3-4 respectively. The final two-hour exam will assess the whole course, but will be weighted heavily towards unit 5.

Team Project: Students will work in teams of three or four students on a realistic and open ended problem on some aspect of climate change. The project will involve literature survey, planning, data analysis/calculations, an oral presentation and written report. Full details of assessment criteria and timetable will be handed out with the project description.

Grades: The overall points for the course will be the weighted average of the points for each assessment item.



COURSE REQUEST SYSTEM - PRODUCTION
ADD REQUEST (Pending)
***** DRAFT *****

11/16/2020 Course: **CHE 150** Version: **1** Title: **Climate Eng for Global Warming**

User Unit Code: **22**

Date Entered: 2/17/2020 1:59:34 PM

by

Last Modified: 11/12/2020 4:21:25 PM

by Alan Zdunek, Director of Undergraduate Studies

General reason for this request:

A General Education course entitled "Climate Engineering for Global Warming" is requested. The course will be under the general education categories of "Analyzing the Natural World" and "Understanding the Past". This course is designed to help students understand the engineering perspective of climate change, its causes, its impact on earth and our society, the risks posed by climate change, and policies to mitigate those risks. Since this is an important topic that the ChE department is addressing, the department wishes to offer the course permanently under its own rubric, number and name.

This request should be processed at the same time as the following related academic program request(s):

None

Requested type of approval and effective date:

Permanent approval effective Fall semester, 2021

Course Subject (Rubric): CHE / Chemical Engineering

Primary Unit: Chemical Engineering

Course Number: 150

Course Version: 1

Course Title: Climate Engineering for Global Warming

Transcript Title: Climate Eng for Global Warming

Course Description:

Overview of the engineering perspective of climate change and strategies to engineer the earth's climate and mitigate risks associate with global warming.

Notes to Students:

None

Cross Listings:

None

Previous Course Number and/or Subject:

None

Expected Registration:

Freshman and sophomore undergraduate

Junior and senior undergraduate

Type of Course:

General Education credit

GENERAL EDUCATION CREDIT

Category

Analyzing the Natural World

General Education Learning Outcomes

- Understand and critically evaluate information and concepts in the natural and mathematical sciences.

Assessment Methods

Rank (if applicable)

1

Written assignment

- Use scientific and mathematical reasoning to make relevant distinctions among ideas.

Assessment Methods

Exam

- Think critically about contemporary issues in science and technology.

Assessment Methods

Project

Understanding the Past

2

General Education Learning Outcomes

- Understand the implication and meaning of technological innovation and scientific discovery for the development of human society.

Assessment Methods

Exam

- Critically analyze the cultural, economic, geographical, and political processes that influenced historical events.
- Understand and explain the significance and influence of the past and its connection to current political, scientific, and cultural forces.

Assessment Methods

Project

Justification for inclusion in the General Education category or categories:

Climate change is a topic of immense and imminent importance. In this regards, This course is designed to help students understand the engineering perspective of climate change, strategies to engineer earth's climate, monitor and predict climate change, and policies to mitigate risks associated with climate change. Through this course, the students will develop an understanding of the various aspects of science, engineering, economics, and policies related to climate change. This course will answer the following questions:

1. How can climate engineering impact global warming and climate change?
2. What is the science behind carbon dioxide removal and albedo modification?
3. What (and how) renewable technologies contribute to decarbonization?
4. How much time will the climate engineering strategies take to impact climate change on the earth?
5. How science, policies, and discussions can inform choices to reduce the risks posed by climate change?

Course Learning Outcomes:

1. Students will be able to understand how climate engineering can impact global warming and climate change.
2. Students will be able to acquire knowledge on the science behind carbon dioxide removal and albedo modification.
3. Students will be able to recognize what renewable technologies contribute to decarbonization.
4. Students will be able to comprehend how science, polices and discussions can inform choices to reduce the risks posed by climate change.
5. Students will be able to understand climate engineering strategies and how long they will take to impact climate change.

Course Learning Outcome Assessment Methods:

- 1) Bi-Weekly homework assignments will assess students' knowledge of course material.
- 2) Two exams and a final exam will test students' understanding of the topics covered in classroom.
- 3) A team project will assess students' knowledge on climate change and global warming concepts and their ability to effectively communicate verbally and in writing.

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

The course does not overlap with any other ChE courses. The course discusses the topics of climate change and global warming which are timely and of interest to students across various disciplines.

Relationship of this course to similar courses offered by other academic units:

N/A

Major Topics:

- 3 hours: Introduction to climate engineering and history of climate change
- 6 hours: Climate Monitoring and Prediction
- 9 hours: Solar Radiation Management (SRM)
- 9 hours: Decarbonization of Energy Systems
- 6 hours: Greenhouse gas removal technologies
- 6 hours: Adapting to climate change- economics and policies
- 3 hours: Exams
- 3 hours: Presentations

TOTAL: 45 hrs

Sample Sources and Resource Materials:

Suggested Text:

- 1) John Houghton, Global Warming: The Complete Briefing, 5th Edition, 2015, Cambridge Univ. Press.
- 2) D. Jacob, Introduction to Atmospheric Chemistry
<http://acmg.seas.harvard.edu/people/faculty/djj/book/index.html>
- 3) Edmond A. Mathez, and Jason E. Smerdon, Climate Change: The Science of Global Warming and our Energy Future, 2nd Edition, 2018, Columbia University Press, NY

Prerequisite(s):

[None](#)

Recommended Background:

[None](#)

Corequisites:

[None](#)

Restrictions:

[None](#)

Credit Restrictions:

[None](#)

Credit Hours:

3 hours.

Type of Instruction:

<i>Type of Instruction</i>	<i>Contact Hours/Week Over 15-Week Term</i>
Lecture	3

Grading Mode: Normal

Sign-Offs:

To be forwarded for review, sign-off, and comment to the following administrators:

Chairperson(s) or director(s) of department(s) or unit(s):

[None](#)

Dean(s) of college(s) or school(s):

[None](#)

Comments related to request for sign-off:

[None](#)

Frequency of Offering:

Offering varies.

Fall or Spring Every Year

Special Technology or Equipment Support Request:

[None](#)

Faculty Proposer(s):Vikas Berry Meenesh Singh

PRIMARY DEPARTMENT APPROVAL STATUS**APPROVED** for Chemical Engineering by Sohail Murad, Head on 2/17/2020 2:18:37 PM**PRIMARY COLLEGE APPROVAL STATUS****NOT YET APPROVED** by Engineering**OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS****NOT YET APPROVED**

COURSE REQUEST SYSTEM - PRODUCTION
CHANGE REQUEST (Pending)
***** DRAFT *******11/16/2020** Course: **CHE 321** Version: **6** Title: **Chemical Reaction Eng.**User Unit Code: **22****Date Entered:** 11/12/2020 4:02:58 PM**by** Alan Zdunek, Director of Undergraduate Studies**Last Modified:** 11/12/2020 4:08:43 PM**by** Alan Zdunek, Director of Undergraduate Studies**General reason for this request:**

Change prerequisite CHE 301 to concurrent registration. Chemical engineering students have been introduced to the principles covered in other chemical engineering and chemistry courses. However, taking CHE 301 concurrently will provide additional depth while learning the CHE 321 material.

This request should be processed at the same time as the following related academic program request(s):

None

Requested type of approval and effective date:

Permanent approval effective Spring semester, 2021

 Previously:

Permanent approval effective Fall semester, 2017

Course Subject (Rubric): CHE / Chemical Engineering**Primary Unit:** Chemical Engineering**Course Number:** 321**Course Version:** 6**Course Title:** Chemical Reaction Engineering**Transcript Title:** Chemical Reaction Eng.**Course Description:**

Kinetics of homogeneous single reactions. Ideal reactors: batch, stirred tank and plug flow systems. Conversion and yield in multiple reactions. Design and optimization of reactors. Non-isothermal reactors.

Notes to Students:

None

Cross Listings:

None

Previous Course Number and/or Subject:

None

Expected Registration:

Junior and senior undergraduate

Type of Course:

Requirement for the following programs: Chemical Engineering

Course Learning Outcomes:

To teach students chemical kinetic rate laws and mechanisms of single versus multiple homogeneous reactions
To introduce analysis and design of batch versus flow reactors
To introduce catalytic reaction engineering and quantify ideal versus real dynamics
To illustrate temperature effects in non-isothermal reactors
To reinforce mathematical analysis and computational skills.
To illustrate the ramifications of societal issues on reactor engineering, including safety

Course Learning Outcome Assessment Methods:

Quizzes, Examinations, Homeworks, Final Examination, and the certificate of completion of a relevant SaChE online module entitled "Runaway Reactions".

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

It is a necessary course in any chemical engineering curriculum.

Relationship of this course to similar courses offered by other academic units:

DNA

Major Topics:

Introduction	2 Hours
Ideal Reactors (batch, CSTR, plug flow)	9 Hours
Reactor design	3 Hours
Interpretation of batch reactor data	3 Hours
Kinetics and mechanisms of homogeneous reactions	3 Hours
Catalysis and Catalytic mechanisms	6 Hours
Catalytic Reactors	4 Hours
Temperature Effects	9 Hours
Introduction to Non Ideal Reactors	6 Hours
Total Hours	45 Hours

Sample Sources and Resource Materials:

"Elements of Chemical Reaction Engineering", 5th Edition, by Fogler, H. Scott, Pearson Education Publication, 2016
Also acceptable: Editions 3 and 4

Prerequisite(s):

CHE 210; and MATH 220; and Credit or concurrent registration in CHE 301.

 Previously:

CHE 210; and MATH 220; and CHE 301.

Recommended Background:

None

Corequisites:

None

Restrictions:

Restricted to students in the following colleges/schools: Engineering or Graduate College.

Credit Restrictions:

None

Credit Hours:

3 hours.

Type of Instruction:

<i>Type of Instruction</i>	<i>Contact Hours/Week Over 15-Week Term</i>
Lecture-Discussion	3

Grading Mode: Normal

Course may be audited by registered visitors.

Sign-Offs:**To be forwarded for review, sign-off, and comment to the following administrators:**

Chairperson(s) or director(s) of department(s) or unit(s):

None

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

Frequency of Offering:

Frequency

Special Notes

Fall of every year.

Spring of every year.

 **Previously:**

No response

Special Technology or Equipment Support Request:

None

Faculty Proposer(s):

Alan Zdunek

PRIMARY DEPARTMENT APPROVAL STATUS

APPROVED for Chemical Engineering by Vikas Berry, Head on 11/12/2020 4:10:08 PM

PRIMARY COLLEGE APPROVAL STATUS

NOT YET APPROVED by Engineering

OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS

NOT YET APPROVED

COURSE REQUEST SYSTEM - PRODUCTION
CHANGE REQUEST (Pending)
***** DRAFT *******11/16/2020** *Course: CHE 396 Version: 3 Title: Senior Design I* *User Unit Code: 22***Date Entered:** 11/12/2020 4:32:27 PM**by** Alan Zdunek, Director of Undergraduate Studies**Last Modified:** 11/12/2020 4:52:50 PM**by** Alan Zdunek, Director of Undergraduate Studies**General reason for this request:**

CHE 396 is the first course of the two semester capstone chemical engineering senior design course sequence (CHE 396/397); recent improvements in the course have decreased the course load/level of effort and justifies a 3-credit hour course.

This request should be processed at the same time as the following related academic program request(s):

Proposed Revision of the BS in Chemical Engineering Program Change: Fall 2020

Requested type of approval and effective date:

Permanent approval effective Fall semester, 2021

 Previously:

Permanent approval effective Spring semester, 2014

Course Subject (Rubric): CHE / Chemical Engineering**Primary Unit:** Chemical Engineering**Course Number:** 396**Course Version:** 3**Course Title:** Senior Design I**Transcript Title:** Senior Design I**Course Description:**

Introduction to modern, process design and development, engineering economics, and report writing. Design and cost of equipment relating to materials handling to heat transfer, mass transfer, and reactors.

Notes to Students:

None

Cross Listings:

None

Previous Course Number and/or Subject:

None

Expected Registration:

Junior and senior undergraduate

Type of Course:

Requirement for the following programs: Chemical Engineering

Course Learning Outcomes:

1. Students will be able to design individual pieces of process equipment, such as heat exchangers, reactors, pumping-piping systems, separation columns.
2. Students will master elements of process economics and analysis of profitability.
3. Students will understand and be able to apply concepts of inherently safer design and control/mitigation of hazards.

Course Learning Outcome Assessment Methods:

Hour exams, homeworks, final exam, presentations, and the certificates of completion of two relevant SaChE online modules: (1) "Inherently Safer Design", and (2) "Safety in the Process Industries".

 Previously:

Hour exams, homeworks, final exam, and the certificates of completion of two relevant SaChE online modules: (1) "Inherently Safer Design", and (2) "Safety in the Process Industries".

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

This is the first in a series of two courses at the senior level aimed at giving senior engineering students some experience and fundamentals in the principles of chemical engineering process design. The course will provide a real-world perspective on process design, introduce design and selection of equipment, give insight into practical constraints that govern design decisions, and improve presentation, analytical thinking and team working skills.

 Previously:

This is the first in a series of two courses at the senior level aimed at giving senior engineering students some experience and fundamentals in the process of chemical engineering design.

Relationship of this course to similar courses offered by other academic units:

DNA

Major Topics:

1. Chemical Design Introduction	3 hours
2. Flowsheet development	6 hours
3. Utilities and Energy Recovery	3 hours
4. Process simulation	6 hours
5. Process control	3 hours
6. Materials of construction	3 hours
7. Capital cost	3 hours
8. Operating cost	3 hours
9. Economic Analysis	3 hours
10. Process Hazard, Professional ethics	3 hours
11. Professional Ethics Presentations	4 hours
12. General Site Consideration	3 hours
13. Exams	2 hours

Total Hours 45 Hours

 Previously:

Introduction to chemical plant design	21 Hours
Optimum design, engineering economics	
How diagrams in relation to production of chemicals-material-energy balances etc.	
Writing of design report	2 Hours
Cost and design of equipment relating of materials transfer and handling	8 Hours
Cost and design of equipment relating to mass transfer	6 Hours
Cost and design of equipment relating to mass transfer and chemical reactors	8 Hours
Small design problems	8 Hours
Statistical analysis as applied to design problems	4 Hours
Concepts of inherently safer design and control/mitigation of hazards	1 Hour
Examinations	2 Hours

Total Hours 60 Hours

Sample Sources and Resource Materials:

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009
Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

 Previously:

*Plant Design and Economics for Chemical Engineers by M.S. Peters & K.D. Timmerhaus, McGraw-Hill Book Co., New York 1980. Material and research articles given and referred to in the class.

Prerequisite(s):

CHE 312 and CHE 313 and CHE 321 ChE 312 (Transport Phenomena II), ChE 313 (Transport Phenomena III) and ChE 321 (Chemical Reaction Engineering).

 Previously:

ChE 312 (Transport Phenomena II), ChE 313 (Transport Phenomena III) and ChE 321 (Chemical Reaction Engineering).

Recommended Background:

None

Corequisites:

None

Restrictions:

Restricted to students in the following colleges/schools: Engineering or Graduate College.

 Previously:

Restricted to students in the following colleges/schools: Engineering or Graduate College or Summer Session Only.

Credit Restrictions:

None

Credit Hours:

3 hours.

 Previously:

4 hours.

Type of Instruction:

Type of Instruction	Contact Hours/Week Over 15-Week Term
Lecture-Discussion	3

 Previously:

Type of Instruction	Contact Hours/Week
Lecture-Discussion	4

Grading Mode: Normal

Course may be audited by registered visitors.

Sign-Offs:

To be forwarded for review, sign-off, and comment to the following administrators:

Chairperson(s) or director(s) of department(s) or unit(s):

None

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

Frequency of Offering:

Frequency

Fall of every year.

Special Notes

 **Previously:**

No response

Special Technology or Equipment Support Request:None

Faculty Proposer(s):

Alan Zdunek Vikas Berry

 **Previously:**None

PRIMARY DEPARTMENT APPROVAL STATUS**APPROVED** for Chemical Engineering by Vikas Berry, Head on 11/12/2020 4:53:58 PM**PRIMARY COLLEGE APPROVAL STATUS****NOT YET APPROVED** by Engineering**OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS****NOT YET APPROVED**

COURSE REQUEST SYSTEM - PRODUCTION
CHANGE REQUEST (Pending)
***** DRAFT *****
11/16/2020 *Course: CHE 397 Version: 5 Title: Senior Design II* *User Unit Code: 22*
Date Entered: 3/11/2020 10:34:43 AM**by** Alan Zdunek, Director of Undergraduate Studies**Last Modified:** 3/11/2020 10:52:19 AM**by** Alan Zdunek, Director of Undergraduate Studies**General reason for this request:**

CHE 397 was made a 3 or 4 variable credit hour course to allow students who had enrolled prior to Fall 2016 to register with 3 credit hours due to credit hour changes in our curriculum in AY 2016/2017. The variable credit hour is no longer needed and can be changed to a flat 4 credit hour course.

This request should be processed at the same time as the following related academic program request(s):

Not applicable

Requested type of approval and effective date:

Permanent approval effective Spring semester, 2021

 Previously:

Permanent approval effective Fall semester, 2016

Course Subject (Rubric): CHE / Chemical Engineering**Primary Unit:** Chemical Engineering**Course Number:** 397**Course Version:** 5**Proposed Course Subject (Rubric) :** CHE / Chemical Engineering**Proposed Primary Unit:** Chemical Engineering**Proposed Course Number:** 397**Course Title:** Senior Design II**Transcript Title:** Senior Design II**Course Description:**

Application of principles and design methodology of Chemical Engineering to the design of large-scale Chemical Processes and Plants. A major design project is assigned for solution and presentation by students working in small groups.

Notes to Students:

Extensive computer use required.

 Previously:

Extensive computer use required. Continuing students will take the course for 3 hours and students entering in Fall 2016 will take the course for 4 hours.

Cross Listings:

None

Previous Course Number and/or Subject:

None

Expected Registration:

Junior and senior undergraduate

Type of Course:

Requirement for the following programs: B.S. in Chem Engineering

Course Learning Outcomes:

To teach students communication skills, oral and written communication, teamwork skills and project management

To teach students the methodology of chemical process and plant

design.

To impart literature-analysis and state-of-art-review skills.

To teach equipment costing and plant economics.

To teach flow charting and materials / energy balance analysis on the plant scale.

To provide facility with process modeling and design software.

To impart equipment specification skills.

To introduce concepts of professional ethics and responsibility.

To illustrate process redesign, analysis of design alternatives and design optimization.

To illustrate environmental impact and energy-related concerns.

To be able to apply basic hazard identification and risk assessment techniques.

Course Learning Outcome Assessment Methods:

Final design project (written report and oral presentation), periodic updates of design project, industrial mentor evaluation, presentation at Engineering Expo, Quizzes, Examinations, Homeworks, Final Examination, the certificate of completion of a relevant SaChE online safety tutorial.

Previously:

Final design project (written report and oral presentation), periodic updates of design project, industrial mentor evaluation, presentation at Engineering Expo, Quizzes, Examinations, Homeworks, Final Examination, and the certificate of completion of a relevant SaChE online module entitled "Risk Assessment".

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

This project-orientated course integrates Chemical Engineering design principles and methodology. It represents the most important design component of accredited Chemical Engineering program.

Relationship of this course to similar courses offered by other academic units:

None

Previously:

DNA

Major Topics:

Process description	5 hours
Design basis, BFD, PFD Flowsheet development	5 hours
Equipment list and sizing	5 hours
Utilities demand summary for entire plant	5 hours
Process simulation	5 hours
Process control strategy, process parameters	5 hours
Materials of construction	5 hours
Plant Cost Analysis (NPV, sensitivity analysis, payback period, capital cost, Aspen economic analysis	5 hours
Environmental and safety concerns	2 hours
Process hazards, mitigation	3 hours
Engineering Design Project	15 Hours

Total 60 Hours

Previously:

Students taking 4 credit hours:

Design Specifications	3 Hours
Literature Search	3 Hours
State of the Art Review	4 Hours
Flow-Charting	3 Hours
Materials and Energy Balance	3 Hours
Equipment Design	5 Hours
Detailed Materials and Engery Balance	4 Hours
Equipment Specification	3 Hours
Equipment Pricing	3 Hours
Plant Optimization	3 Hours
Economics: Overall Pricing and Forecasting	4 Hours
Professional ethics and responsibility	2 Hours
Pinch analysis for energy integration	1 Hour
Environmental impact and EPA regulations	1 Hour

Process redesign	1 Hour	
Alternative design analysis	1 Hour	
Hazard identification and risk assessment	1 Hour	
Engineering Design Project	15 Hours	
		Total 60 Hours
Students taking 3 credit hours:		
Design Specifications	3 Hours	
Literature Search	3 Hours	
State of the Art Review	4 Hours	
Flow-Charting	3 Hours	
Materials and Energy Balance	3 Hours	
Equipment Design	5 Hours	
Detailed Materials and Engery Balance	4 Hours	
Equipment Specification	3 Hours	
Equipment Pricing	3 Hours	
Plant Optimization	3 Hours	
Economics: Overall Pricing and Forecasting	4 Hours	
Professional ethics and responsibility	2 Hours	
Pinch analysis for energy integration	1 Hour	
Environmental impact and EPA regulations	1 Hour	
Process redesign	1 Hour	
Alternative design analysis	1 Hour	
Hazard identification and risk assessment	1 Hour	
		Total 45 Hours

Sample Sources and Resource Materials:

1. Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012

a. Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker, 2018.

 Previously:

1. "Plant Design and Economics for Chemical Engineers" (3rd Edition) by M.S. Peters and K.D. Timmerhaus; McGraw-Hill, New York (1980).
2. "Chemical Engineers" Handbook, 5th Ed., by R.H. Perry and C.H. Chilton (Eds.); Reference Texts.

Prerequisite(s):

CHE 396.

Recommended Background:

None

Corequisites:

None

Restrictions:

Restricted to students in the following colleges/schools: Engineering or Graduate College.

 Previously:

Restricted to students in the following colleges/schools: Engineering or Graduate College or Summer Session Only.

Credit Restrictions:

None

Credit Hours:

4 hours.

 Previously:

3 to 4 hours.

Type of Instruction:

Type of Instruction

Lecture-Discussion

Contact Hours/Week Over 15-Week Term

4

Continuing students will take the course for 3 hours and students entering the degree in Fall 2016 will take the course for 4 hours.

Grading Mode: Normal

Course may be audited by registered visitors.

Sign-Offs:**To be forwarded for review, sign-off, and comment to the following administrators:**

Chairperson(s) or director(s) of department(s) or unit(s):

None

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

Frequency of Offering:

Frequency

Special Notes

Spring of every year.

Special Technology or Equipment Support Request:

ASPEN Plus simulation software is required for design analysis

Faculty Proposer(s):

Alan Zdunek

PRIMARY DEPARTMENT APPROVAL STATUS

APPROVED for Chemical Engineering by Vikas Berry, Head on 11/12/2020 11:38:44 AM

PRIMARY COLLEGE APPROVAL STATUS

NOT YET APPROVED by Engineering

OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS

NOT YET APPROVED

COURSE REQUEST SYSTEM - PRODUCTION
CHANGE REQUEST (Pending)
***** DRAFT *******11/16/2020** Course: **CHE 514** Version: **2** Title: **Biotransport**User Unit Code: **22****Date Entered:** 9/24/2020 5:42:28 PM**by** Monica Soto-Malfavon**Last Modified:** 11/12/2020 11:36:55 AM**by** Alan Zdunek, Director of Undergraduate Studies**General reason for this request:**

To switch control from CHE 514 to BIOE 514.

This request should be processed at the same time as the following related academic program request(s):

None

Requested type of approval and effective date:

Permanent approval effective Spring semester, 2021

 Previously:

Permanent approval effective Summer semester, 1993

Course Subject (Rubric): CHE / Chemical Engineering **Proposed Course Subject (Rubric) :** BIOE / Bioengineering**Primary Unit:** Chemical Engineering**Proposed Primary Unit:** Bioengineering**Course Number:** 514**Proposed Course Number:** 514**Course Version:** 2**Course Title:** Biotransport**Transcript Title:** Biotransport**Course Description:**

Diffusion and flow in living systems. Blood rheology and flow. Microcirculation, oxygen transport, diffusive transport across membranes. Membrane structure; water, and ion flows, active transport.

Notes to Students:

None

Cross Listings:**CHE 514/**Chemical Engineering /Chemical Engineering

It is a standard course elective toward a graduate degree in both BioE and ChE.

Percentage of instruction provided by controlling department or comments regarding shared instruction:
100%, not shared**Controlling course:** CHE 514/Chemical Engineering/ Chemical Engineering **Previously:****BIOE 514/**Bioengineering /Bioengineering

It is a standard course elective toward a graduate degree in both BioE and ChE.

Percentage of instruction provided by controlling department or comments regarding shared instruction:
100%, not shared**Controlling course:** CHE 514/Chemical Engineering /Chemical Engineering**Previous Course Number and/or Subject:**

None

Expected Registration:

Graduate College

Type of Course:

Selective for the following programs: MS and PHD

 Previously:

Selective for the following programs:

Course Learning Outcomes:

Students will be able to:

1. Analyze and examine diffusion and flow in living systems.
2. Discuss, evaluate and investigate diffusion-reaction in biological and biomedical engineering systems.
3. Discuss blood rheology and flow.
4. Analyze and study microcirculation, oxygen transport, and diffusive transport across membranes.
5. Classify, critique and discuss steady state and dynamics of biomedical engineering processes vital to the analysis and design of medical devices for artificial clinical intervention.

 Previously:

No response

Course Learning Outcome Assessment Methods:

Homeworks, Exams, Project

 Previously:

No response

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

Advanced course in transport phenomena area of concentration of graduate programs in Chemical Engineering and Bioengineering.

 Previously:

Advanced course in transport phenomena area of concentration of graduate programs in Chemical Engineering and Bioengineering.

Relationship of this course to similar courses offered by other academic units:

DNA

Major Topics:

I. Flow Processes		
A. The cardiovascular system	4	
B. Rheology of blood and other biofluids		4
C. Arterial blood flow	4	
D. Venous blood flow	2	
E. Microcirculation	6	
F. Oxygen transport - lungs and tissue		4
G. Tracer analysis of flow	4	
II. Diffusive Processes		
A. Membrane structures	4	
B. Theories of diffusion	4	
C. Ion and solute transport	4	
D. Water transport	4	
E. Active transport	4	
F. Process models	4	
Student Presentations	8	
	Total 60	

Sample Sources and Resource Materials:

Current literature in biomedical and biochemical engineering.

Prerequisite(s):

BIOE 325 (or equivalent) or consent of the instructor.

 Previously:

ChE 410 Transport Phenomena, or consent of the instructor.

Recommended Background:

None

Corequisites:

None

Restrictions:

None

Credit Restrictions:

None

Credit Hours:

4 hours.

Type of Instruction:

Type of Instruction

Contact Hours/Week Over 15-Week Term

Lecture-Discussion

4

Grading Mode: Normal

Course may be audited by registered visitors.

Sign-Offs:**To be forwarded for review, sign-off, and comment to the following administrators:**

Chairperson(s) or director(s) of department(s) or unit(s):

2203 Bioengineering

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

 **Previously:**

Chairperson(s) or director(s) of department(s) or unit(s):

Dean(s) of college(s) or school(s):

Frequency of Offering:

No response

Special Technology or Equipment Support Request:

None

Faculty Proposer(s):

None

PRIMARY DEPARTMENT APPROVAL STATUS**APPROVED** for Chemical Engineering by Vikas Berry, Head on 11/12/2020 11:37:13 AM**SIGNOFF STATUS****SIGNED OFF** for Bioengineering by Thomas Royston, Head on 11/12/2020 11:41:09 AM**Comments:**

Approved per TR

PRIMARY COLLEGE APPROVAL STATUS**NOT YET APPROVED** by Engineering**OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS****NOT YET APPROVED**

<i>Title:</i>	Proposal to Revise the Bachelor of Science in Computer Engineering											
<i>Sponsor:</i>	Department of Electrical and Computer Engineering College of Engineering											
<i>Executive Summary:</i>	The Department of Electrical and Computer Engineering (ECE) proposes to add the following courses to the list of CE Technical Electives 1. Add ECE 454 Mechatronics Embedded Systems Design 2. Add ECE 464 Testing and Reliability of Digital Systems 3. Add ECE 491 Seminar The proposed change keeps the number of hours required for graduation to 128.											
<i>Detailed Description of Proposed Changes and Justification:</i>	The ECE department offers three majors: Electrical Engineering (EE), Computer Engineering (CE) and Engineering Physics (EP). Both the EE and CE majors include 17 hours of “Technical elective” courses, higher level courses of direct relevance to the degree. Students may select courses from a list of approved technical elective courses. We are proposing to add two new courses to this list for our Bachelor of Science in Computer Engineering. These courses are new technical courses in our department and are at the 400 level, perfectly suited to higher level undergraduates.											
<i>Justification:</i>	See above.											
<i>Catalog Statement:</i>	The following table summarizes the degree requirements with the present curriculum (right-hand side) and with the proposed curriculum (left-hand side). A detailed catalog statement can be found in Appendix. <table><tr><td>Present Summary of CE Requirements</td><td>NO CHANGES</td></tr><tr><td>Nonengineering and General Education Requirements 53</td><td></td></tr><tr><td>Required in the College of Engineering 58</td><td></td></tr><tr><td>Technical Electives 17</td><td></td></tr><tr><td>Total Hours 128</td><td></td></tr></table>		Present Summary of CE Requirements	NO CHANGES	Nonengineering and General Education Requirements 53		Required in the College of Engineering 58		Technical Electives 17		Total Hours 128	
Present Summary of CE Requirements	NO CHANGES											
Nonengineering and General Education Requirements 53												
Required in the College of Engineering 58												
Technical Electives 17												
Total Hours 128												
<i>Minority Impact Statement:</i>	There should be no impact on minority students at UIC.											

<i>Budgetary and Staff Implications:</i>	No impact is anticipated because no new courses are created (the new courses are already being offered, they are simply not on the technical elective list).
<i>Library Resource Implications:</i>	No impact is anticipated because courses are neither created nor eliminated.
<i>Space Implications:</i>	No impact is anticipated because no new courses are created.
<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	Approved by: Electrical and Computer Engineering, full faculty, on October 27, 2020. College of Engineering, Educational Policy Committee, on ??? 2020.
<i>Contact Person:</i>	(Curriculum Coordinator) Natasha Devroye, devroye@uic.edu, SEO 1039, phone 6.1013. (Department Head) Daniela Tuninetti, danielat@uic.edu, SEO 1027, phone 3.7431.
<i>Proposed Effective Date/Term:</i>	Spring 2021.

Appendix A – Catalog for BS in Electrical Engineering

Present catalog, on the left-hand side, from <https://catalog.uic.edu/ucatalog/colleges-depts/engineering/ece/bs-ce/> Proposed catalog on the right-hand side.

BS in Computer Engineering Degree Requirements To earn a Bachelor of Science in Computer Engineering degree from UIC, students need to complete university, college, and department degree requirements. The Department of Electrical and Computer Engineering degree requirements are outlined below. Students should consult the <i>College of Engineering</i> section for additional degree requirements and college academic policies. See the ECE Department website for any revisions of the CE curriculum www.ece.uic.edu .	BS in Computer Engineering Degree Requirements SAME
Summary of Requirements Nonengineering and General Education Requirements 53 Required in the College of Engineering 58 Technical Electives 17 Total Hours 128	Summary of Requirements SAME
Nonengineering and General Education Requirements Required Courses ENGL 160 Academic Writing I: Writing in Academic and Public Contexts 3 ENGL 161 Academic Writing II: Writing for Inquiry and Research 3 Exploring World Cultures course ^a 3 Understanding the Creative Arts course ^a 3 Understanding the Past course ^a 3 Understanding the Individual and Society course ^a 3	Nonengineering and General Education Requirements SAME SAME SAME SAME SAME

Understanding U.S. Society course ^a	3	SAME
MATH 180 Calculus I ^b	4	SAME
MATH 181 Calculus II ^b	4	SAME
MATH 210 Calculus III ^b	3	SAME
MATH 220 Introduction to Differential Equations	3	SAME
MATH 310 Applied Linear Algebra.	3	SAME
PHYS 141 General Physics I (Mechanics) ^b	4	SAME
PHYS 142 General Physics II (Electricity and Magnetism) ^b	4	SAME
PHYS 260 Introduction to Thermal Physics.	2	SAME
CHEM 122 General Chemistry I Lecture ^c	4	SAME
CHEM 123 General Chemistry Laboratory I ^{b,c}	1	SAME
Total Hours	53	SAME
a: Students should consult the General Education section of the catalog for a list of approved courses in this category.		SAME
b: This course is approved for the Analyzing the Natural World General Education category.		SAME
c: General Education credit is given for successful completion of both CHEM 122 and CHEM 123.		
Required in the College of Engineering		Required in the College of Engineering
Required Courses		
ENGR 100 Engineering Orientation ^a	1	SAME
CS 107 Introduction to Computing and Programming	4	SAME
CS 151 Mathematical Foundations of Computing	3	SAME
CS 251 Data Structures	4	SAME
ECE 115 Introduction to Electrical and Computer Engineering	4	SAME

ECE 225 Circuit Analysis	4	SAME
ECE 265 Introduction to Logic Design	4	SAME
ECE 266 Introduction to Embedded Systems	4	SAME
ECE 310 Discrete and Continuous Signals and Systems	3	SAME
ECE 333 Computer Communication Networks	4	SAME
ECE 340 Electronics I	4	SAME
ECE 341 Probability and Random Processes for Engineers	3	SAME
ECE 366 Computer Organization	3	SAME
ECE 396 Senior Design I	2	SAME
ECE 397 Senior Design II	2	SAME
ECE 465 Digital Systems Design	3	SAME
ECE 466 Advanced Computer Architecture.	3	SAME
ECE 467 Introduction to VLSI Design	4	SAME
ECE 499 Professional Development Seminar	0	SAME
Total Hours	58	Total Hours 58
a: ENGR 100 is a one-semester-hour course, but the hour does not count toward the total hours required for graduation.		
Technical Electives		Technical Electives
No more than a total of two courses below the 400-level may be used to meet the technical elective requirement. Also, at most one course from outside the Electrical and Computer Engineering Department may be used to meet the technical elective requirement ^a . Select 17 hours from the following:		SAME
PHYS 240 Fundamentals of Modern Quantum Theory		SAME
CS 361 Computer Systems		SAME
CS 401 Computer Algorithms I		SAME
ECE 311 Communication Engineering		SAME
ECE 317 Digital Signal Processing I		SAME

ECE 322 Communication Electromagnetics	SAME
ECE 342 Electronics II	SAME SAME
ECE 346 Solid State Device Theory	SAME
ECE 347 Integrated Circuit Engineering	
ECE 350 Principles of Automatic Control	SAME
ECE 401 Quasi-Static Electric and Magnetic Fields	SAME
ECE 407 Pattern Recognition I	SAME
ECE 410 Network Analysis	SAME
ECE 412 Introduction to Filter Synthesis	SAME
ECE 415 Image Analysis and Computer Vision I	SAME
ECE 417 Digital Signal Processing II	SAME
ECE 418 Statistical Digital Signal Processing	SAME
ECE 421 Introduction to Antennas and Wireless Propagation	SAME
ECE 423 Electromagnetic Compatibility	SAME
ECE 424 RF and Microwave Guided Propagation	SAME
ECE 427 Modern Linear Optics	SAME
ECE 431 Analog Communication Circuits	SAME
ECE 432 Digital Communications	SAME
ECE 434 Multimedia Systems	SAME
ECE 436 Computer Communication Networks II	SAME
ECE 437 Wireless Communications	SAME
	SAME
ECE 442 Power Semiconductor Devices and Integrated Circuits	SAME
ECE 445 Analysis and Design of Power Electronic Circuits	SAME
ECE 448 Transistors	SAME
ECE 449 Microdevices and Micromachining Technology	SAME
ECE 451 Control Engineering	

	ADD NEW COURSE: ECE 454 Mechatronic Embedded Systems Design
ECE 452 Robotics: Algorithms and Control	SAME
ECE 458 Electromechanical Energy Conversion	SAME
	ADD NEW COURSE: ECE 464 Testing and Reliability of Digital Systems
ECE 468 Analog and Mixed - Signal VLSI Design	SAME
ECE 469 Computer Systems Design	SAME
	ADD NEW COURSE: ECE 491 Seminar (these courses are used for new special topics courses – we put titles then -- and should be counted as technical electives)
MCS 425 Codes and Cryptography	SAME
MCS 471 Numerical Analysis	SAME
STAT 471 Linear and Non-Linear	SAME
^a <i>Students preparing for the Fundamentals of Engineering Examination, which leads to becoming a Licensed Professional Engineer, are advised to use these hours to take <u>CME 201</u>, and one of the following to prepare for the Fundamentals of Engineering Exam: <u>CME 203</u>, <u>CME 260</u>, or <u>ME 211</u>.</i>	

<i>Title:</i>	Proposal to Revise the Bachelor of Science in Electrical Engineering											
<i>Sponsor:</i>	Department of Electrical and Computer Engineering College of Engineering											
<i>Executive Summary:</i>	The Department of Electrical and Computer Engineering (ECE) proposes to add the following courses to the list of EE Technical Electives 1. Add ECE 454 Mechatronics Embedded Systems Design 2. Add ECE 464 Testing and Reliability of Digital Systems 3. Add ECE 491 Seminar The proposed change keeps the number of hours required for graduation to 128.											
<i>Detailed Description of Proposed Changes and Justification:</i>	The ECE department offers three majors: Electrical Engineering (EE), Computer Engineering (CE) and Engineering Physics (EP). Both the EE and CE majors include 17 hours of “Technical elective” courses, higher level courses of direct relevance to the degree. Students may select courses from a list of approved technical elective courses. We are proposing to add two new courses to this list for our Bachelor of Science in Electrical Engineering. These courses are new technical courses in our department and are at the 400 level, perfectly suited to higher level undergraduates.											
<i>Justification:</i>	See above.											
<i>Catalog Statement:</i>	The following table summarizes the degree requirements with the present curriculum (right-hand side) and with the proposed curriculum (left-hand side). A detailed catalog statement can be found in Appendix. <table><tr><td>Present Summary of EE Requirements</td><td>NO CHANGES</td></tr><tr><td>Nonengineering and General Education Requirements 53</td><td></td></tr><tr><td>Required in the College of Engineering 58</td><td></td></tr><tr><td>Technical Electives 17</td><td></td></tr><tr><td>Total Hours 128</td><td></td></tr></table>		Present Summary of EE Requirements	NO CHANGES	Nonengineering and General Education Requirements 53		Required in the College of Engineering 58		Technical Electives 17		Total Hours 128	
Present Summary of EE Requirements	NO CHANGES											
Nonengineering and General Education Requirements 53												
Required in the College of Engineering 58												
Technical Electives 17												
Total Hours 128												
<i>Minority Impact Statement:</i>	There should be no impact on minority students at UIC.											

<i>Budgetary and Staff Implications:</i>	No impact is anticipated because no new courses are created (the new courses are already being offered, they are simply not on the technical elective list).
<i>Library Resource Implications:</i>	No impact is anticipated because courses are neither created nor eliminated.
<i>Space Implications:</i>	No impact is anticipated because no new courses are created.
<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	Approved by: Electrical and Computer Engineering, full faculty, on October 27, 2020. College of Engineering, Educational Policy Committee, on ??? 2020.
<i>Contact Person:</i>	(Curriculum Coordinator) Natasha Devroye, devroye@uic.edu, SEO 1039, phone 6.1013. (Department Head) Daniela Tuninetti, danielat@uic.edu, SEO 1027, phone 3.7431.
<i>Proposed Effective Date/Term:</i>	Spring 2021.

Appendix A – Catalog for BS in Electrical Engineering

Present catalog, on the left-hand side, from <https://catalog.uic.edu/ucat/colleges-depts/engineering/ece/bs-ee/>. Proposed catalog on the right-hand side.

BS in Electrical Engineering Degree Requirements To earn a Bachelor of Science in Electrical Engineering degree from UIC, students need to complete university, college, and department degree requirements. The Department of Electrical and Computer Engineering degree requirements are outlined below. Students should consult the <i>College of Engineering</i> section for additional degree requirements and college academic policies. See the ECE Department website for any revisions of the EE curriculum www.ece.uic.edu.	BS in Electrical Engineering Degree Requirements SAME												
Summary of Requirements <table> <tr> <td>Nonengineering and General Education Requirements</td><td>53</td></tr> <tr> <td>Required in the College of Engineering</td><td>58</td></tr> <tr> <td>Technical Electives</td><td>17</td></tr> <tr> <td>Total Hours</td><td>128</td></tr> </table>	Nonengineering and General Education Requirements	53	Required in the College of Engineering	58	Technical Electives	17	Total Hours	128	Summary of Requirements SAME				
Nonengineering and General Education Requirements	53												
Required in the College of Engineering	58												
Technical Electives	17												
Total Hours	128												
Nonengineering and General Education Requirements Required Courses <table> <tr> <td>ENGL 160 Academic Writing I: Writing in Academic and Public Contexts</td><td>3</td></tr> <tr> <td>ENGL 161 Academic Writing II: Writing for Inquiry and Research</td><td>3</td></tr> <tr> <td>Exploring World Cultures course ^a</td><td>3</td></tr> <tr> <td>Understanding the Creative Arts course ^a</td><td>3</td></tr> <tr> <td>Understanding the Past course ^a</td><td>3</td></tr> <tr> <td>Understanding the Individual and Society course ^a</td><td>3</td></tr> </table>	ENGL 160 Academic Writing I: Writing in Academic and Public Contexts	3	ENGL 161 Academic Writing II: Writing for Inquiry and Research	3	Exploring World Cultures course ^a	3	Understanding the Creative Arts course ^a	3	Understanding the Past course ^a	3	Understanding the Individual and Society course ^a	3	Nonengineering and General Education Requirements SAME SAME SAME SAME SAME
ENGL 160 Academic Writing I: Writing in Academic and Public Contexts	3												
ENGL 161 Academic Writing II: Writing for Inquiry and Research	3												
Exploring World Cultures course ^a	3												
Understanding the Creative Arts course ^a	3												
Understanding the Past course ^a	3												
Understanding the Individual and Society course ^a	3												

Understanding U.S. Society course ^a	3	SAME
MATH 180 Calculus I ^b	4	SAME
MATH 181 Calculus II ^b	4	SAME
MATH 210 Calculus III ^b	3	SAME
MATH 220 Introduction to Differential Equations	3	SAME
MATH 310 Applied Linear Algebra.	3	SAME
PHYS 141 General Physics I (Mechanics) ^b	4	SAME
PHYS 142 General Physics II (Electricity and Magnetism) ^b	4	SAME
PHYS 260 Introduction to Thermal Physics.	2	SAME
CHEM 122 General Chemistry I Lecture ^c	4	SAME
CHEM 123 General Chemistry Laboratory I ^{b,c}	1	SAME
Total Hours	53	SAME
a: Students should consult the General Education section of the catalog for a list of approved courses in this category. b: This course is approved for the Analyzing the Natural World General Education category. c: General Education credit is given for successful completion of both CHEM 122 and CHEM 123.		
Required in the College of Engineering		Required in the College of Engineering
Required Courses		
ENGR 100 Engineering Orientation ^a	1	SAME
CS 107 Introduction to Computing and Programming	4	SAME
ECE 115 Introduction to Electrical and Computer Engineering	4	SAME
ECE 225 Circuit Analysis	4	SAME
ECE 265 Introduction to Logic Design	4	SAME

ECE 266 Introduction to Embedded Systems	4	SAME
ECE 310 Discrete and Continuous Signals and Systems	3	SAME
ECE 311 Introduction to Analog and Digital Communications	4	SAME
ECE 317 Digital Signal Processing I	4	SAME
ECE 322 Introduction to Electromagnetics and Applications	4	SAME
ECE 340 Electronics I	4	SAME
ECE 341 Probability and Random Processes for Engineers	3	SAME
ECE 342 Electronics II	4	SAME
ECE 346 Solid State Device Theory	4	SAME
ECE 350 Principles of Automatic Control	4	SAME
ECE 396 Senior Design I	2	SAME
ECE 397 Senior Design II	2	SAME
ECE 499 Professional Development Seminar	0	SAME
Total Hours	58	Total Hours
a: ENGR 100 is a one-semester-hour course, but the hour does not count toward the total hours required for graduation.		58
Technical Electives		Technical Electives
No more than a total of two courses below the 400-level may be used to meet the technical elective requirement. Also, at most one course from outside the Electrical and Computer Engineering Department may be used to meet the technical elective requirement ^a . Select 17 hours from the following:		SAME
<u>PHYS 240</u>	Fundamentals of Modern Quantum Theory	SAME
<u>ECE 333</u>	Computer Communication Networks I	SAME

<u>ECE 347</u>	Integrated Circuit Engineering	SAME
<u>ECE 366</u>	Computer Organization	SAME SAME
<u>ECE 407</u>	Pattern Recognition I	SAME
<u>ECE 410</u>	Network Analysis	SAME
<u>ECE 412</u>	Introduction to Filter Synthesis	SAME
<u>ECE 415</u>	Image Analysis and Computer Vision I	SAME SAME
<u>ECE 417</u>	Digital Signal Processing II	
<u>ECE 418</u>	Statistical Digital Signal Processing	SAME
<u>ECE 421</u>	Introduction to Antennas and Wireless Propagation	SAME SAME SAME
<u>ECE 423</u>	Electromagnetic Compatibility	SAME
<u>ECE 424</u>	RF and Microwave Guided Propagation	SAME SAME SAME
<u>ECE 427</u>	Modern Linear Optics	SAME
<u>ECE 431</u>	Analog Communication Circuits	SAME SAME
<u>ECE 432</u>	Digital Communications	SAME
<u>ECE 434</u>	Multimedia Systems	SAME SAME
<u>ECE 436</u>	Computer Communication Networks II	
<u>ECE 437</u>	Wireless Communications	SAME
<u>ECE 440</u>	Nanoelectronics	

<u>ECE 442</u>	Power Semiconductor Devices and Integrated Circuits	SAME
<u>ECE 445</u>	Analysis and Design of Power Electronic Circuits	SAME
<u>ECE 448</u>	Transistors	SAME
<u>ECE 449</u>	Microdevices and Micromachining Technology	SAME
<u>ECE 451</u>	Control Engineering	SAME
<u>ECE 452</u>	Robotics: Algorithms and Control	SAME
<u>ECE 458</u>	Electromechanical Energy Conversion	ECE 454 Mechatronics Embedded Systems Design 4 ADD NEW COURSE SAME ECE 464 Testing and Reliability of Digital Systems 3 ADD NEW COURSE
<u>ECE 465</u>	Digital Systems Design	SAME
<u>ECE 466</u>	Advanced Computer Architecture	SAME
<u>ECE 467</u>	Introduction to VLSI Design	SAME
<u>ECE 468</u>	Analog and Mixed - Signal Integrated Circuits	SAME
<u>ECE 469</u>	Hardware Description Language Based Digital and	SAME

Computer System Design		ECE 491 Seminar [1-4 hours] ADD NEW COURSE (these courses are used for new special topics courses – we put titles then -- and should be counted as technical electives)
<u>MCS 425</u>	Codes and Cryptography	
<u>MCS 471</u>	Numerical Analysis	
<u>STAT 471</u>	Linear and Non-Linear Programming	
Total Hours	17	
^a Students preparing for the Fundamentals of Engineering Examination, which leads to becoming a Licensed Professional Engineer, are advised to use these hours to take <u>CME 201</u> , and one of the following to prepare for the Fundamentals of Engineering Exam: <u>CME 203</u> , <u>CME 260</u> , or <u>ME 211</u> .		SAME



COURSE REQUEST SYSTEM - PRODUCTION
ADD REQUEST (Pending)
***** DRAFT *****

11/16/2020 Course: **BIOE 538** Version: **1** Title: **Numerical Analysis** User Unit Code: **22**

Date Entered: 11/20/2019 4:52:33 PM

by Susan Lee, Student Program Coordinator

Last Modified: 2/15/2020 4:04:11 PM

by John R. Hetling, Assistant Professor

General reason for this request:

New permanent course

This request should be processed at the same time as the following related academic program request(s):

None

Requested type of approval and effective date:

Permanent approval effective Fall semester, 2020

Course Subject (Rubric): **BIOE** / Bioengineering

Primary Unit: Bioengineering

Course Number: 538

Course Version: 1

Course Title: Engineering Numerical Analysis and Multiscale Modeling

Transcript Title: Numerical Analysis

Course Description:

Present numerical approaches to solve several canonical systems of ordinary and partial differential equations in engineering. Advanced study of multiscale modeling techniques.

Notes to Students:

Extensive computer use required.

Cross Listings:

None

Previous Course Number and/or Subject:

None

Expected Registration:

Graduate College

Type of Course:

Elective

Course Learning Outcomes:

Students will be able to construct a mathematical model of a biological-physical system and to ensure that it is well-posed.

Students will be able to approximate total and partial derivatives on grids using 1st and 2nd order approximations.

Students will be able to convert equations described by approximate derivatives into systems of algebraic equations and solve them using 1-step or iterative methods.

Students will be able to assess the accuracy of a solution using grid refinement and comparisons to analytical solutions.

Course Learning Outcome Assessment Methods:

Weekly assignments, quiz, and mid-term and final exams will be conducted to assess the students' understanding of the fundamentals of numerical analysis.

Matlab coding assignments will be used to assess the programming ability of the students.

A final short paper and an oral presentation by each student will be used to assess their understanding of the material and to communicate it through writing and verbal methods.

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

Computational modeling is the third mode of discovery in science and engineering, besides experiments and theory. The purpose of this course is to cover the fundamentals and important topics in computational modeling. This course will also prepare students to take more specialized modeling classes in Bioengineering.

Relationship of this course to similar courses offered by other academic units:

ME 428 Numerical Methods in Mechanical Engineering is an undergraduate-level course, while this course (BIOE538/ME538) is at the graduate-level. ME 528 Numerical Heat Transfer, ME 534 Finite Element Analysis II, and ME 542 Advanced Computational Methods for Product and Process Design are specialized modeling courses on heat transfer, solid mechanics, and design, while this course (BIOE538/ME538) focuses on numerical solutions of classical differential equations, regardless of the engineering disciplines. It emphasizes the fundamentals of numerical analysis such as interpolation, integration, and stability analysis rather than a particular method.

Major Topics:

1. Introduction to PDE's and their classification (4 hours)
2. Numerical interpolation (4 hours)
3. Numerical differentiation (4 hours)
4. Numerical integration (4 hours)
5. Backward and forward Euler integration in time (4 hours)
6. Steady state solutions to Laplace's equation in 2-D (4 hours)
7. Hyperbolic PDE's in 1-D space and time (4 hours)
8. Parabolic PDE in 1-D (4 hours)
9. Parabolic PDE in 2-D space and time (4 hours)
10. Hyperbolic-Parabolic PDEs (4 hours)
11. Finite difference methods (4 hours)
12. Introduction to finite element methods (4 hours)
13. All-atoms molecular dynamics (4 hours)
14. Coarse-grained molecular dynamics (4 hours)
15. Multiscale modeling in bioengineering (4 hours)

Total: 60 hours

Sample Sources and Resource Materials:

Moin, P., Fundamentals of Engineering Numerical Analysis, Cambridge University Press, 2nd edition, 2010
 Pozrikidis, C., Numerical Computation in Science and Engineering. Oxford University Press. 2nd edition, 2008.

Prerequisite(s):

Graduate standing.

Recommended Background:

Vector Calculus; Mechanics or Transport; and working knowledge of Matlab, or C, or Fortran programming.

Corequisites:

None

Restrictions:

None

Credit Restrictions:

None

Credit Hours:

4 hours.

Type of Instruction:

Type of Instruction
 Lecture-Discussion

Contact Hours/Week Over 15-Week Term
 3

Grading Mode: Normal

Course may be taken Credit/No Credit by students who qualify for this option.

Sign-Offs:

To be forwarded for review, sign-off, and comment to the following administrators:

Chairperson(s) or director(s) of department(s) or unit(s):

None

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

Frequency of Offering:

Frequency

Special Notes

Spring of every year.

Special Technology or Equipment Support Request:

Some computer lab hours using Linux systems are expected.

Faculty Proposer(s):

Zhangli Peng

PRIMARY DEPARTMENT APPROVAL STATUS

APPROVED for Bioengineering by Thomas Royston, Head on 11/25/2019 1:34:46 PM

Comments:

Approved on behalf of TR

PRIMARY COLLEGE APPROVAL STATUS

NOT YET APPROVED by Engineering

OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS

NOT YET APPROVED



COURSE REQUEST SYSTEM - PRODUCTION
ADD REQUEST (Pending)
***** DRAFT *******11/16/2020****Course: CS 532 Version: 1 Title: Topics in NLP****User Unit Code: 22****Date Entered:** 10/26/2020 1:19:51 PM**by** Santhi Nannapaneni, Assistant Director for Student Affairs**Last Modified:** 10/26/2020 1:25:21 PM**by** Santhi Nannapaneni, Assistant Director for Student Affairs

General reason for this request:

The purpose of this request is to create a permanent graduate-level course that covers advanced contemporary research topics in natural language processing.

This request should be processed at the same time as the following related academic program request(s):[None](#)

Requested type of approval and effective date:

Permanent approval effective Fall semester, 2021

Course Subject (Rubric): CS / Computer Science**Primary Unit:** Computer Science**Course Number:** 532**Course Version:** 1

Course Title: Advanced Topics in Natural Language Processing**Transcript Title:** Topics in NLP

Course Description:

Investigation of advanced contemporary topics in natural language processing, including analyses of research principles, methodologies, and evaluation techniques. Course topics evolve in concert with key advances and trending themes in NLP research.

Notes to Students:[None](#)

Cross Listings:[None](#)

Previous Course Number and/or Subject:[None](#)

Expected Registration:

Graduate College

Type of Course:

Elective

Course Learning Outcomes:

Students will be able to independently analyze modern scholarship in emerging topics in natural language processing research; understand and communicate advanced ideas and methodologies; and identify areas of improvement to pursue as follow-up research. Students will also be able to perform their own original research where these contemporary topics intersect with their own interests.

Course Learning Outcome Assessment Methods:

Written paper critiques will assess students' abilities to develop an understanding of advanced research methodologies, identify key advantages and limitations of the presented techniques, and analyze whether the paper's conclusions are justified.

Oral paper presentations will assess students' abilities to identify the most important aspects of research papers, and clearly communicate advanced concepts at a level accessible to other members of the research community.

Course projects, project reports, and project presentations will assess students' abilities to apply the methodologies discussed in class to make novel contributions to contemporary research areas in natural language processing.

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

The field of natural language processing is experiencing unprecedented growth, and that growth is accompanied by an exciting set of emerging (and rapidly evolving) research topics. The purpose of this course is to familiarize students with key contemporary advancements in natural language processing, leaving them well-equipped to pursue advanced follow-up research on their own or to be attractive candidates for industry roles requiring these skillsets. Students will also develop a better understanding of the current research landscape, strengthen their abilities to conduct thoughtful critical analyses of trending literature, and build expertise in key methods, evaluation techniques, and popular NLP tools.

This course will also allow UIC's Department of Computer Science to showcase its strong presence in the natural language processing community. By establishing a permanent advanced natural language processing course as part of the curriculum and course catalog, prospective and current students will develop a better understanding of the research breadth and specific interests of natural language processing faculty. This exposure may also lead to increased interest from high-caliber prospective graduate students, further strengthening the department's presence in this area. Finally, this will allow us to overcome the constraints of repeated use of CS 594 for NLP topics courses, all of which have in recent years been met with strong student interest and high enrollment numbers.

Relationship of this course to similar courses offered by other academic units:

The course overlaps in topic area with CS 421, CS 521, CS 582, and CS 583. While those core courses provide foundational knowledge necessary to develop an understanding of contemporary and fundamental research, their emphasis is on building a working understanding of standard principles, algorithms, and other methodologies and terminology common in the field. In contrast, the proposed course will focus on critically analyzing and developing an in-depth understanding of specific facets of contemporary work that may not yet be covered in more standardized curriculum, or may require substantial background knowledge.

Major Topics:

1. Contemporary fundamentals of NLP, computer vision, and deep learning (9 hours)
2. Project Proposals (3 hours)
3. Grounded Language Learning (6 hours)
4. Physically Situated Dialogue (4.5 hours)
5. Multimodal Affective Analysis (1.5 hours)
6. Automated Image Captioning and Video Description (3 hours)
7. Image-Text Alignment (3 hours)
8. Multimodal Story Entailment (1.5 hours)
9. Project Updates (3 hours)
10. Visual Question Answering (1.5 hours)
11. Multimodal Language Disambiguation (3 hours)
12. Project Presentations (6 hours)

Total: 45 hours

Sample Sources and Resource Materials:

Research papers from:

- Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)
- Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)
- Proceedings of the International Conference on Computational Linguistics (COLING)
- Proceedings of the Conference on Computational Natural Language Learning (CoNLL)
- Computational Linguistics Journal
- Transactions of the Association for Computational Linguistics (TACL)

Prerequisite(s):

CS 421 or CS 521 or CS 582 or CS 583; or consent of the instructor.

Recommended Background:

At least one natural language processing course (e.g., CS 421), and at least one AI or machine learning course (e.g., CS 411 or CS 412).

Corequisites:

None

Restrictions:

Restricted to students in the following colleges/schools: Graduate College.

Credit Restrictions:

None

Credit Hours:

4 hours.

Type of Instruction:

Type of Instruction	Contact Hours/Week Over 15-Week Term
Lecture-Discussion	3

Grading Mode: Normal**Sign-Offs:****To be forwarded for review, sign-off, and comment to the following administrators:**

Chairperson(s) or director(s) of department(s) or unit(s):

None

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

Frequency of Offering:

Offering varies.

Special Technology or Equipment Support Request:

None

Faculty Proposer(s):

Natalie Parde Barbara Di Eugenio Cornelia Caragea Bing Liu

PRIMARY DEPARTMENT APPROVAL STATUS

APPROVED for Computer Science by Robert Sloan, Head on 10/26/2020 1:52:52 PM

Comments:

Approved by GC committee 5/15/2020 per DGS (BDE).

PRIMARY COLLEGE APPROVAL STATUS

NOT YET APPROVED by Engineering

OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS

NOT YET APPROVED



COURSE REQUEST SYSTEM - PRODUCTION
ADD REQUEST (Pending)
***** DRAFT *****

11/17/2020 Course: **CS 351** Version: **1** Title: **Adv. Data Structure Practicum** User Unit Code: **22**

Date Entered: 10/2/2020 10:19:05 AM

by Santhi Nannapaneni, Assistant Director for Student Affairs

Last Modified: 11/16/2020 11:53:28 AM

by Santhi Nannapaneni, Assistant Director for Student Affairs

General reason for this request:

The content of this course covers design and implementation details in advanced data structures and practical algorithms. The content covers complex data structures and algorithms with emphasis on implementation details and run-time analysis.

This request should be processed at the same time as the following related academic program request(s):

[None](#)

Requested type of approval and effective date:

Permanent approval effective Spring semester, 2021

Course Subject (Rubric): **CS** / Computer Science

Primary Unit: Computer Science

Course Number: 351

Course Version: 1

Course Title: Advanced Data Structure Practicum

Transcript Title: Adv. Data Structure Practicum

Course Description:

Design and implementation details of advanced data structure and non-trivial algorithms with an emphasis on amortized analysis.

Notes to Students:

[None](#)

Cross Listings:

[None](#)

Previous Course Number and/or Subject:

[None](#)

Expected Registration:

Junior and senior undergraduate

Type of Course:

Elective

Course Learning Outcomes:

Students will be able to implement various complex data structures from scratch and use the said data structures to solve problems.

Students will be able to apply different algorithm design techniques to solve computing problems efficiently.

Course Learning Outcome Assessment Methods:

The ability of students to understand complex data structures will be assessed by several projects and assignments where they are required to implement said data structures and use them to solve problems.

The ability of students to understand algorithm design techniques will be assessed by peer interviews on a biweekly basis.

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

Expands student knowledge on data structures by introducing complex data structures and focusing on implementation details and amortized analysis.
Covers popular algorithm design techniques with focus on implementation.

Relationship of this course to similar courses offered by other academic units:

This course is an extension to CS 251 and a prelude to CS 401.

Major Topics:

- Overview of basic data structures (4 hours of lecture/discussion, 8 hours of student-directed preparation)
- AVL trees, RedBlack trees, Tries (6 hours of lecture/discussion, 10 hours of student-directed preparation)
- Binary, Binomial, and Fibonacci heaps (6 hours of lecture/discussion, 10 hours of student-directed preparation)
- Disjoint sets (2 hours of lecture/discussion, 4 hours of student-directed preparation)
- Graph algorithms (5 hours of lecture/discussion, 12 hours of student-directed preparation)
- Greedy Approach (4 hours of lecture/discussion, 8 hours of student-directed preparation)
- Dynamic Programming (5 hours of lecture/discussion, 10 hours of student-directed preparation)
- Network Flow (4 hours of lecture/discussion, 8 hours of student-directed preparation)
- Miscellaneous (string algorithm, bitwise manipulation, computational geometry) (4 hours of lecture/discussion, 8 hours of student-directed preparation)

Total hours = 40 lecture and discussion hours + 78 hours of student-directed activities (assignments, projects, review material) = 118

Sample Sources and Resource Materials:

"Introduction to Algorithms" by Cormen, Leiserson, Rivest, Stein.

Prerequisite(s):

Credit or concurrent registration in CS 251.

Recommended Background:

None

Corequisites:

None

Restrictions:

Restricted to students in the following majors: Computer Science: BS.

Credit Restrictions:

None

Credit Hours:

3 hours.

Type of Instruction:

<i>Type of Instruction</i>	<i>Contact Hours/Week Over 15-Week Term</i>
Lecture-Discussion	3

Grading Mode: Normal

Sign-Offs:

To be forwarded for review, sign-off, and comment to the following administrators:

Chairperson(s) or director(s) of department(s) or unit(s):

None

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

Frequency of Offering:

Offering varies.

Special Technology or Equipment Support Request:

None

Faculty Proposer(s):

Nasim Mobasheri

PRIMARY DEPARTMENT APPROVAL STATUS

APPROVED for Computer Science by Robert Sloan, Head on 11/16/2020 11:53:59 AM

Comments:

Approved by ugrad committee 11/11/2020.

PRIMARY COLLEGE APPROVAL STATUS

NOT YET APPROVED by Engineering

OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS

NOT YET APPROVED



COURSE REQUEST SYSTEM - PRODUCTION
ADD REQUEST (Pending)
***** DRAFT *******11/17/2020** Course: **CS 378** Version: **1** Title: **Framework-based app developmen** User Unit Code: **22****Date Entered:** 9/17/2020 11:44:25 AM**by** Santhi Nannapaneni, Assistant Director for Student Affairs**Last Modified:** 10/1/2020 12:56:15 PM**by** Santhi Nannapaneni, Assistant Director for Student Affairs

General reason for this request:

The course addresses two emerging Computer Science trends, namely (1) framework-based software development and (2) app development for hand-held devices. To date, new software products are typically developed on top of existing frameworks and libraries thereby reducing cost and time-to-market. This trend is especially evident in mobile app development because frameworks allow for developing apps simultaneously for different platforms such as Android and iOS. This course is exclusively focused on these two topics.

This request should be processed at the same time as the following related academic program request(s):

Change in title of CS 478

Requested type of approval and effective date:

Permanent approval effective Summer semester, 2021

Course Subject (Rubric): CS / Computer Science**Primary Unit:** Computer Science**Course Number:** 378**Course Version:** 1

Course Title: Framework-based software development for hand-held devices**Transcript Title:** Framework-based app developmen

Course Description:

Multi-platform app development using software frameworks; the integration of framework-generated code with native code of mobile operating systems; numerous programming assignments in multiple programming languages.

Notes to Students:

None

Cross Listings:

None

Previous Course Number and/or Subject:

None

Expected Registration:

Junior and senior undergraduate

Type of Course:

Elective

Course Learning Outcomes:

1. Students will be able to design and implement medium-sized mobile apps for at least two different hand-held device platforms.
 2. Students will understand how to integrate existing software frameworks with application-specific code in an effort to develop new products.
 3. Students will learn how to develop software using frameworks.
 4. Students will be able to assess the effort involved in developing a mobile app.
-

Course Learning Outcome Assessment Methods:

1. Numerous software projects will test student knowledge on how to integrate frameworks

with application-specific code and how to design and implement mobile apps for at least two software platforms.

2. In class exams and quizzes will assess student's ability to develop software using frameworks and to understand the effort involved.

Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:

The course will introduce students to framework-based software development; this topic is not currently addressed systematically in the current CS curricula. In addition, the course will cover in detail software development for hand-held devices, which are prevalent to date. There is a minor overlap with CS 478, which covers native app development in a given platform. CS 478 does not cover framework-based multi-platform development; instead, it emphasizes the relations between app development and other areas of CS.

Relationship of this course to similar courses offered by other academic units:

None known.

Major Topics:

1. Survey of popular software frameworks, their architecture and the Model-View-Controller design pattern. Examples of such frameworks include Spring, Rails and React. (1 week)
2. Survey programming language for chosen framework for mobile app development (e.g., Dart for Flutter, JavaScript for React Native) (3 weeks)
3. Detailed coverage of one framework for mobile app development, including framework deployment, app development and review of generated app code (5 weeks)
4. Extending framework-generated code with native iOS code or newer OS (3 weeks)
5. Extending framework-generated code with native Android code or newer OS (2 weeks)
6. In class exams and quizzes (1 week)

Total - 15 weeks

Sample Sources and Resource Materials:

Methods and tools for app development evolve so frequently that books are often obsolete by the time they hit bookshelves. Excellent resources are available on the web, however.

Here are some samples:

<https://developer.android.com/>

<https://developer.apple.com/>

<https://flutter.dev/>

and many more.

Prerequisite(s):

CS 342.

Recommended Background:

None

Corequisites:

None

Restrictions:

Restricted to students in the following colleges/schools: Engineering.

Credit Restrictions:

None

Credit Hours:

3 hours.

Type of Instruction:

Type of Instruction

Lecture-Discussion

Contact Hours/Week Over 15-Week Term

3

Grading Mode: Normal

Sign-Offs:

To be forwarded for review, sign-off, and comment to the following administrators:*Chairperson(s) or director(s) of department(s) or unit(s):*

None

Dean(s) of college(s) or school(s):

None

Comments related to request for sign-off:

None

Frequency of Offering:*Frequency**Special Notes*

Fall of every year.

Special Technology or Equipment Support Request:

None

Faculty Proposer(s):

Ugo Buy and Mark Hallenbeck

PRIMARY DEPARTMENT APPROVAL STATUS**APPROVED** for Computer Science by Robert Sloan, Head on 11/16/2020 12:00:04 PM**Comments:**

Approved by ugrad committee - 10/2/20.

PRIMARY COLLEGE APPROVAL STATUS**NOT YET APPROVED** by Engineering**OFFICE OF VICE CHANCELLOR FOR ACADEMIC AFFAIRS APPROVAL STATUS****NOT YET APPROVED**

EPC meeting #3 Fall 2020

Thursday, 12/17/2020, 3-5 pm

Present: Ataei, Darabi, DiEugenio (for Item 1), Devroye, Hetling, Khodadoust, Kotche, Muench, Troy, Zdunek

Meeting Minutes

1. CS 532 – Professor DiEugenio. Sample listing in major topics updated to better reflect learning outcomes. **Unanimous approval by EPC, updated in CRS.**
2. CME BS in Environmental Engineering – Professor Khodadoust. First submission of proposal to EPC. Expands environ engr minor. CME professional advisory board strongly endorses. 128 credit hours enforced, not requiring initial new course development to support. Aiming for Fall 2022. **Unanimous approval by EPC, sent to Dana.**
3. CS 351. Modified CRS submission based on EPC feedback in previous meeting. **Unanimous approval by EPC.**
4. CS 378. Modified CRS submission based on EPC feedback in previous meeting. **Unanimous approval by EPC.**
5. BIOE 538 tabled until next meeting
6. CHE 396/397 Prereqs approved and faculty approval date updated. Updated CRS submission based on feedback from previous EPC meeting. **Unanimous approval by EPC.**
7. CHE 427. Moving from CHE 494 special topics course. Modifying title “Entrepreneurship in Engineering”, description and major topics to be more broad for other COE majors (remove “chemical”). Consider cross listed in ENGR. Pre-req: Junior and senior standing. **Unanimous approval by EPC with aforementioned changes.**
8. CHE 514. Move course to BIOE 514 “Advanced Biotransport. **Unanimous approval by EPC.**

Topics:

1. Standing meetings for 2021 <https://academicprograms.uic.edu/timeline/>
Miiri will send a Doodle poll for spring 2021 monthly meetings
2. ABET
 - Degree audit?
 - Formal way to get feedback on program educational objectives from judges via Engr EXPO survey for judges and career fair participants.
3. 194 / 294 / 394 /494
CS implemented, ECE in process of submitting for new classes.
“design spine” throughout the curriculum

EPC meeting #1 Spring 2021

Friday, 1/22/2020, 10-11:20am

Present: Ataei, Darabi, Devroye, Hetling, Khodadoust, Kotche, Muench, Troy, Zdunek

Meeting Minutes

1. SCEP meeting 1/14: approved B.S. in CS+Design [no action required](#)
2. CME 508: lab component, prereqs [approved unanimously by EPC](#)
3. ME 396/397, co-req. (396 not approved by dept) [approved 397 unanimously by EPC, Jim deleted 396 from CRS.](#)
4. CHE 101/ CME 260 / CME 261 / CME 322 / ME 261

CHEM 112 prereq to be updated to CHEM 122. [depts will submit CRS request, EPC voted to approve based on this administrative change](#)

5. BIOE courses >> BME courses deadline 2/15/21 [EPC voted to approve based on this administrative change as they come in queue](#)
6. In CRS queue for COE approval:

CHE 427 (approved last EPC meeting) – [Jim approved](#)

CS 351 (approved last EPC meeting) – [Jim approved](#)

ECE 407 (cross listing with BIOE, waiting for BIOE approval) – [Miiri to email Susan Lee](#)

What are these? MENG 414/415/432/465/466/475 (pending from 2017) – [Miiri to print and send to George and Carolyn, see if they can be deleted from CRS](#)

7. BIOE 562 / ME 562 to be cross listed CME 562 – [Jim will confirm with Dana that a new class doesn't have to be created](#)

Topics:

- Advising process – iAdvise?

[DUGS agreed to pilot iAdvise this semester \(along with any other interested faculty advisors\). UA needs to give access to DUGS, list as secondary advisor for list of students who will be advised \(list to be requested from undergrad coordinators\), in order to roll out pilot. UA will provide training/instructions for pilot. Pilot trial week 7 or week 8. Include undergrad coordinators for iAdvise access and training. \(CS not included bc their advisors already using iAdvise\)](#)

EPC meeting #2 Spring 2021

Friday, 2/19/2020, 10am-11:30am

Present: Ataei, Darabi, Devroye, Hetling, Kotche, Muench, Troy, Zdunek

Meeting Minutes

1. Approval of Dec and Jan meeting minutes [Approved by EPC](#)
2. iAdvise pilot [Miiri to set up demo next week. Faculty will pilot week 9 or 10.](#)
3. BIOE 526 to be cross listed with CME 526 (currently already x listed with ME 526) [John will initiate](#)
4. BME 194, 294, 394, 494: [Approved by EPC](#)
5. BME 442: [Approved by EPC](#)
6. BME 538: New permanent course [Approved by EPC after update of contact hours](#)
7. CS488: New permanent course [Approved by EPC](#)
8. [John will review all other BIOE course changes to present for March EPC](#)
9. MIE 499 – how do other depts handle prereqs? [It's a universal challenge. Transfer articulation is the primary issue with current system.](#)
10. CME 222 prereq – can they be different for CHE and CME students? [Cross list the class between departments or call out prereqs for each dept \(but then enforcement is a concern\)](#)
11. ENG 289, 489 [possibly can be removed from CRS?](#)
12. IDS 573, IE 201, 464, 596, IPD 501, 502, 510 521, 522, 523, 524, 531, 532, 533, ME 503, 516, 596 [Houshang will look into if these courses should be deleted from CRS](#)
13. MENG 514, 515, 432. 465, 466, 475 – [Miiri to print and send to George and Carolyn, see if they should be deleted from CRS](#)

EPC meeting #4 Spring 2021

Friday, 4/16/2020, 10am-12pm

Present: Hossein Atai, Houshang Darabi, Natasha Devroye, John Hetling, Miiri Kotche, Pat Troy, Alan Zdunek

Meeting Minutes

CRS submissions:

- ECE 210: change to the prereqs **comprehensive rationale in EPC folder meeting minutes, unanimously approved**
- CME 260, CME 261/ME 261: change to the prereqs **update to reflect new CHEM course numbers, unanimously approved**
- BIOE 475 – COE can't approve without x-listing with BIOS **John will follow up with BIOS DUGS**
- CS 453: new course "Introduction to Parallel and Distributed Processing" **Unanimously approved pending minor rephrasing of learning outcomes**
- CS 454: new course "Principles of Concurrent Programming" **Unanimously approved**
- CS 483: new course "Big Data Mining" **Update to remove CME 594 (Intro to Data Science) in reference to similar courses offered by other academic units. Miiri to follow up with Jim if there are other considerations for not referencing 494/594 courses.**
- CS 516: new course "Responsible Data Science and Algorithmic Fairness" **Unanimously approved**
- CS 518: new course "Deep Learning for Computer Vision" Request to expand course description. **Include ECE 415/515 Deep Learning courses in "Purpose of this course in relation to other courses offered by other units"**
- BME 479: new course "wearables tech lab" **Prereqs for CS majors: Pat recs CS 362 or CS 211/261 for CS majors, since cross-listed; waiting for review in CS Grad Committee meeting. Unanimously approved pending finalization of CS prereqs.**
- BME 538: new course, "numerical analysis" **Course language will be revised to emphasize BME applications.**

Other Topics for discussion:

- Prereq check logic being built into Banner – 100/200 level first, focus on MIE and CME, completion goal for SPR22. Currently coded in as of 4/9/21 in advance of registration by Dana Wright:
 - **CHE 201**
 - CME 201*, CME 203*, CME 205, CME 211, CME 260, CME 261 / ME 261, CME 290
 - CS 107, CS 109
 - ECE 210*, ECE 225*
 - IE 201
 - ME 205, ME 210*, ME 211
 - **previous prereq checks*
- Fall 2021 effective date for incoming freshmen and transfers: CHE 101, CME 112, CS 141, CS 151, and ECE 115
- Remaining (200-level ?) courses to be ready for Spring 2022
- Prerequisite overrides move to departments
 - The Banner screen is SZACSRP, override code is PREREQ.
- iAdvise update:
 - Feedback?
 - Share at Admin Council meeting
 - Summer/fall transition for all faculty
 - Point person in dept? Trainings?
 - Staff – spreadsheet to lift holds, not emails from students

Houshang raised concern that course advising holds are in free text notes rather than discrete fields. MIE dept still sends additional spreadsheet with each course in separate columns. Query-based tracking and analysis is prevented with this method as enrollment grows. Long term request – can drop down menu be integrated with course listing?

- GPPA to PhD program: <https://gppa.uic.edu/programs-offered/>

Committee discussed how little GPPA program was utilized for PhD studies, and idea of modifying GPPA for MS program was raised. Miiri to discuss with DGS.